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Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

EPIDURAL BLOOD PATCHING

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1. Introduction and Purpose

Epidural blood patching (EBP) is a technique used to treat post dural puncture headache (PDPH). It is used once conservative treatment (involving bed rest, giving fluid and simple analgesics) has failed. It involves injecting autologous blood into the epidural space to seal the hole in the dura. It has a 95% immediate success rate which dwindles to around 70% over time. Up to 30% of patients may therefore need a second blood patch. However, it is associated with risks such as persistent back, leg and neck pains.

2. Scope

This guideline is intended as a reference for non-obstetric Consultant Anaesthetists seeking a recommended technique for epidural blood patching.

3. Points to consider

3.1 Patient Preparation

- Procedure to be performed by duty Consultant Anaesthetist and preferably undertaken during daytime hours
- Full assessment of the patient especially including a detailed history of the headache, whether postural and if associated with any signs of cranial nerve palsies, followed by a clinical examination.
- Indications for epidural blood patch (EBP) include :
    i. Post dural puncture headache not resolving following 24 hours of conservative treatment
    ii. Post dural puncture headache associated with deafness or signs of other cranial nerve palsies (urgent indication)
- Ensure no contraindication to epidural blood patch :
    i. Patient refusal
    ii. Thrombocytopaenia (platelet count < 80)
    iii. Coagulopathy (INR >1.4) or patient on anticoagulation therapy
    iv. Local or systemic sepsis

- Discussion with the patient of:

    a) Benefits of an epidural:

        i. Instant relief of PDP headache

        ii. 95% immediate success rate for treating PDPH (70% longer term success rate following single EBP)

    b) Risks of an epidural blood patch :

        i. Backache (can last up to 1 month)

        ii. Neck pain

        iii. Pain in back of legs

        iv. Failure of procedure (with or without further dural punctures) or failure of epidural blood patch itself once in situ (30% longer term failure rate of EBP (following a successful EBP procedure) thus requiring a second epidural blood patch)

        v. Nerve damage (1:24,000 incidence at 6 months)

    c) Epidural blood patch procedure

    d) Explanation of what sensations patient might feel when blood is injected into the epidural space (ie. back pain and posterior leg pain)

- Obtain written consent for the epidural blood patch procedure

3.2 Equipment and Staff

- Perform procedure in theatre

- This should be a two person procedure so ensure another medical practitioner is available (to take the blood sample) in addition to the ODP to give assistance during the epidural blood patch procedure

- Check all equipment is present to perform epidural blood patch (NRFit butterfly device for venesection is available on the epidural trolley)

3.3 Technique

- Site a **16G** (or 14G) iv cannula if not already in situ

- Commence a crystalloid infusion if not already present

- Place patient preferably in the **lateral** position so as to reduce the symptoms of headache. Alternatively, perform it with the patient sitting

- Both operators should use full aseptic technique :

  a) Gown, mask and gloves

  b) Spray the patient's back twice from the mid thoracic area to the upper natal cleft, and over to the flanks using **Derma Spray** (0.5% chlorhexidine in 70% alcohol). Allow the alcohol to dry each time

- Skin and ligament infiltration with 1% Lignocaine **preferably at one lumbar interspace below the site of the original dural puncture otherwise at the same level as the dural puncture**

- 16G Tuohy needle (8cm) used and inserted at this lumbar interspace

- Epidural space is located using loss of resistance to saline technique

- The scrubbed medical assistant should then clean the antecubital fossa using 0.5% chlorhexidine in 70% alcohol and drape the area before proceeding to remove 40 – 45 mls of blood aseptically using an NRFit butterfly device attached to a 20ml NRFit syringe

- 20-25 mls of the blood should be passed to the anaesthetist performing the blood patch who will then inject this volume slowly 5 ml at a time via the Tuohy needle into the epidural space

- Stop injecting and remove the Tuohy needle either when :

  i. The patient gets significant back or leg pain that does not resolve within a few seconds and returns immediately on renewed injection

  ii. A maximum of 25 mls of blood has been given

- Inject the remaining 20 mls of the blood sample into aerobic and anaerobic blood culture bottles for microscopy, culture and sensitivity analysis

- Transfer the patient back to the ward and keep supine for 2 hours following the epidural blood patch procedure

- If the patient is headache free on mobilising following this 2 hour period then she can be discharged home if otherwise fit for discharge

- She should be given instructions to avoid lifting and straining for 1week and return to the hospital if the headache recurs, or visual or hearing disturbances develops

- If the headache returns following a period of mobilisation then the patient should be offered a second blood patch

- If symptoms persist after a second epidural blood patch then reconsider your diagnosis. A CT scan may need to be ordered to exclude other rarer causes of headache eg. cortical vein thrombosis, subdural haematoma, subarachnoid haemorrhage and brain tumour.

4. References

1. Postoperative complications associated with regional anaesthesia in the parturient. Norris M.C. (2000) *Handbook of Obstetric Anaesthesia. First Edition.* Lippincott Williams & Wilkins : 546 – 552